Experiment 8.2

Shielding Effectiveness and EMC Compliance Analysis using MATLAB

Aim

To compute the shielding effectiveness (SE) of a metallic shield over 1 MHz to 1 GHz and to check EMC compliance of the measured emissions against the limit.

Software Required

MATLAB (MATLAB Online)

Procedure / Algorithm

- Define the frequency range 1 MHz – 1 GHz and the copper shield parameters (conductivity 5.8×10^{7} S/m, thickness 1 mm).
- Calculate the skin depth and the absorption loss A.
- Calculate the reflection loss R and the multiple-reflection loss B.
- Obtain the total SE = R + A + B.
- Compare the measured emission level with the EMC limit and display the compliance status.

Output

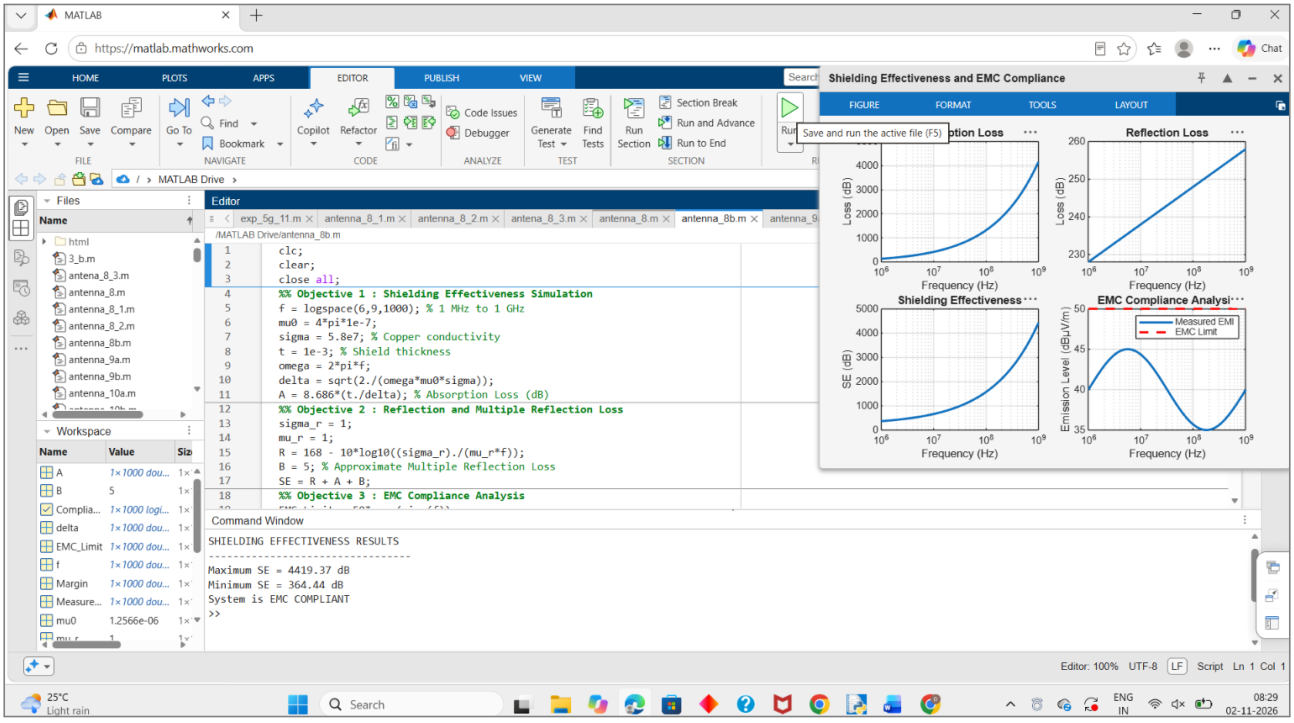


Fig. 8.2 Absorption loss, reflection loss, shielding effectiveness and EMC compliance analysis

Result

- Maximum SE = 4419.37 dB and minimum SE = 364.44 dB over the frequency range.
- The measured emission stays below the EMC limit, so the system is EMC compliant.
- Shielding effectiveness and EMC compliance were analysed successfully.